E2 Observation and Validation Plan

MATH 574, Team 2 · Fahad Bin Gias · 31 October 2026

We fit weekly symptom onsets, times a reporting fraction φ, to San Juan's 2010/11 DENV-1 cases on a square-root scale. We validate with a rolling holdout inside that season and a 2012/13 check, because 2024 is not comparable enough for numbers. One season of cases pins down only m·s0 and φ·s0, so s0 has to come from outside the case data.

What we fit to

The target is `denv1_attributed_cases` in file 02: San Juan's weekly lab-positive cases attributed to DENV-1, 30 Apr 2010 to 29 Apr 2011 (52 weeks). The other two columns are sensitivity checks.

Series (file 02 column)	What it counts	2010/11 total	Role
denv1_attributed_cases	DENV-1 cases, plus untyped and estimated untested cases split by that week's serotype mix	3,864	Primary
denv1_cases	Cases typed as DENV-1 by RT-PCR	2,594	Sensitivity
total_cases	All lab-positive cases plus 22 estimated among untested specimens, all serotypes	4,696	Sensitivity

- Typed-only counts bend the curve. The share of positives that got a serotype ran from 0.55 to 0.94 across weeks with 20 or more cases, lower around the peak than in the tail.
- All-serotype counts mix in DENV-4, 16.7% of typed cases, which a one-serotype model cannot represent.
- Source: the CDC and Puerto Rico Department of Health files on GitHub, already in our data package. The DengAI files on DrivenData or Kaggle will not work: San Juan's labelled weeks stop at 22 Apr 2008, the 2010/11 weeks sit in their unlabelled test set, and there are no serotype columns (DengAI split).

From the model to weekly cases

Expected reported cases in week w are φ times the new symptom onsets in that week:

$$\mu_w = \varphi \int_{t_w}^{t_{w+1}} \epsilon_h E_h(t)\, dt$$

- **Onsets, not infections or prevalence.** The CDC counts are by week of symptom onset, which is the E_h → I_h flow. Infections (S_h → E_h) happen about 6 days earlier, and I_h counts people sick at one moment, not new cases.
- **Real weeks.** t_w comes from `days_since_start`, so 24–31 Dec 2010 stays an 8-day week instead of being squeezed into 7.
- **Exact weekly totals.** A counter state C with dC/dt = ε_h E_h is solved alongside the model; each week's onsets are C(t_{w+1}) − C(t_w).
- **φ counts all infections.** The model has no asymptomatic class, so φ is the share of all DENV-1 infections that end up as a lab-positive case in our data.

$$\varphi = P(\text{symptoms} \mid \text{infected}) \times P(\text{reported} \mid \text{symptoms}) \times P(\text{counted lab-positive} \mid \text{reported})$$

The proposal's 0.035–0.09 is the middle factor: in the 1990s, 10–27 cases went unreported for every case reported to passive surveillance (Noyd and Sharp). The first factor is about 1 in 4 across serotypes (Bhatt et al. 2013) and nearer 1 in 12 for DENV-1 in one outbreak study (Wang et al. 2015). The last is at most 1, since surveillance takes suspected cases and the lab misses some. So φ per infection is roughly 0.003–0.023, lower if the lab misses many reported cases.

Starting state. The season opens mid-epidemic, with 40 DENV-1 cases in week 1. The infected compartments therefore start on the model's own growth mode (the leading eigenvector of the linearised infected block), scaled so weeks 1–3 match the data. The old start, with no infected mosquitoes, makes a fake dip in week 2:

Week	Data	Old start	Growth-mode start
1	39.7	50.7	39.1
2	49.8	38.1	46.9
3	52.4	53.1	56.0
4	56.5	68.0	66.5

The starting size then needs no fitted parameter, so E1 keeps to 2–3 free parameters. E1 can still fit it instead (`onset_rate0`).

Residuals

E1 fits on the square-root scale, where a quiet week and a peak week carry about the same noise. The residual is data minus model, so a positive value means the model is too low:

$$r_w = \sqrt{y_w} - \sqrt{\mu_w}$$

A model-free check on all 23 seasons (1,196 weeks; 1,314 usable points from the all-serotype and DENV-1 series together) measured the week-to-week noise directly from the data. Its variance is 1.2–2.2 times the mean, only mildly noisier than Poisson: a negative binomial size k of roughly 150–300 depending on the series (about 230 for lab-positive counts alone). The table compares the noise variance in weeks with 80+ cases against weeks with 5–20 cases on each scale.

Scale	Noise variance, 80+ vs 5–20 case weeks	Use
Raw counts	20×	No: the peak weeks would decide the fit
Square root	1.5×	Yes: E1's least squares
Negative binomial stabilised (k = 290)	1.1×	Optional: barely better than square root
Log	0.14×	No: the quiet tail would decide the fit

For likelihood work later (MCMC, intervals), use a negative binomial and estimate k along with the other parameters. The noise alone suggests k ≈ 290, but model error will pull it lower.

Diagnostics to report in Stage 3:

- residuals plotted against week and against the fitted value
- lag-1 autocorrelation, and a runs test on the residual signs
- Pearson dispersion with k = 290; near 1 means the model fits as well as the noise allows

A preview fit with provisional parameters gives residual variance 0.69, against 0.30–0.50 from noise alone, and Pearson dispersion 3.7: model error, not reporting noise, dominates. Least-squares standard errors must therefore use the residual variance (2·cost/(n − p) ≈ 0.73), not the noise level. With lag-1 autocorrelation of 0.2, a block bootstrap is safer still.

Residual diagnosis module

`diagnostics.fit_report(res, fun=resid)` takes E1's least-squares result and reports RSS, AIC, AICc and BIC, standard errors, the parameter correlation matrix and the Jacobian's condition number. It flags every pair of parameters with |correlation| ≥ 0.95, meaning the data cannot tell them apart. It recomputes the Jacobian by central differences, because least_squares' default step is smaller than the ODE solver's own error: on a synthetic fit the returned Jacobian hid the s0–m–φ ridge completely (condition number 7 instead of 948). E1 should also fit with diff_step=1e-4.

Fit	Free parameters	RSS	AIC	Condition number	Largest |correlation|
Synthetic, s0 fixed at 0.4	m, φ	22.92	−38.6	1.7	0.19
Synthetic, all free	s0, m, φ	22.91	−36.6	948	1.000, all 3 pairs flagged
Real 2010/11, s0 fixed at 0.4	m, φ	36.47	−14.4	1.7	0.18
Real 2010/11, all free	s0, m, φ	31.43	−20.2	46	≥ 0.996, all 3 pairs flagged

AIC compares fits to the same weeks on the same residual scale. On synthetic data the extra parameter buys nothing and AIC rises by 2, as it should. On the real season the free fit slides along the ridge to s0 = 0.04, m = 90, φ = 0.038 and AIC falls by 5.7, but every pair is flagged and the gain comes from the tail misfit described in the next section, so it is not evidence for that s0.

What one season can pin down

Only two combinations: m·s0 and φ·s0. Halve s0 and double m, and the epidemic grows at the same rate but infects half as many people; double φ as well, and the reported counts are the same.

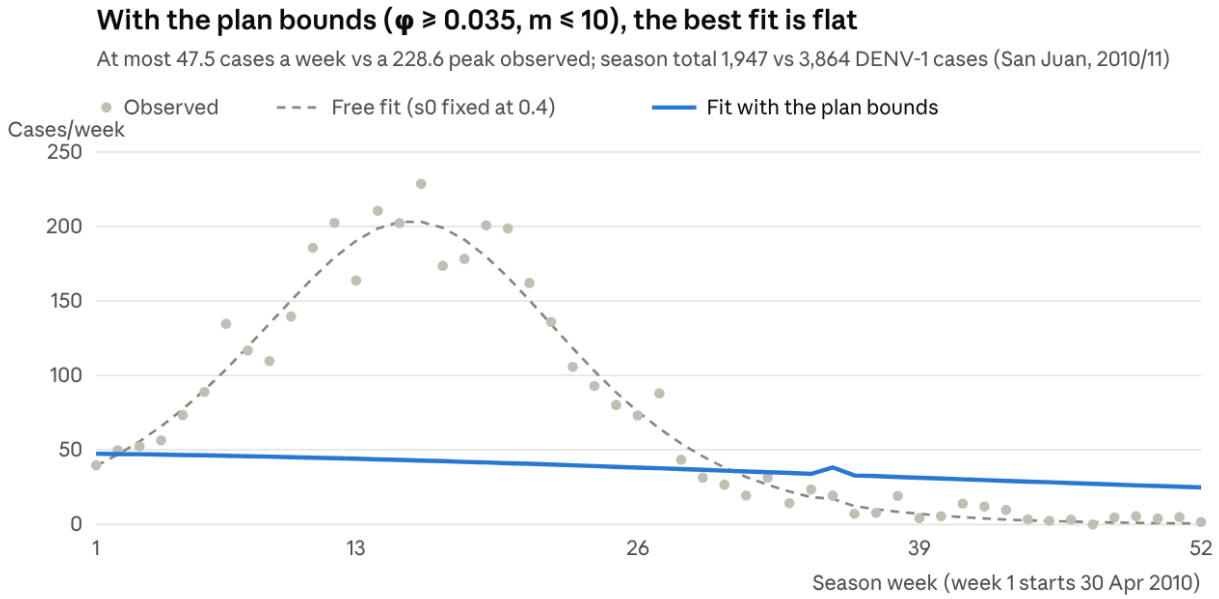
$$(s_0,\ m,\ \varphi) \ \to\ (s_0/\lambda,\ \lambda m,\ \lambda\varphi) \quad \text{leaves } \mu_w \text{ almost unchanged}$$

Only births and the 2% of mosquitoes infected at the peak break this. On the real season, fixing s0 and fitting m and φ gives:

s0 fixed	m	φ	R0	R_eff	Fit cost
0.05	78.3	0.0336	6.44	1.44	15.8
0.1	39.0	0.0181	4.55	1.44	16.9
0.2	19.5	0.0094	3.22	1.44	17.8
0.4	9.8	0.0048	2.28	1.44	18.2
0.6	6.6	0.0032	1.86	1.44	18.3
1.0	4.0	0.0020	1.46	1.46	18.3

Fit cost (half the sum of squared residuals) changes far less than R0, which runs from 6.4 to 1.5. Cost does fall as s0 shrinks, because births then top up a tiny susceptible pool and lift the tail, where the model is too

low (weeks 41–52: 66 cases observed, 26 fitted). That pull comes from a missing mechanism, not from information about s0, so it is not credible; a free fit of s0 would simply run to its lower bound. On fake data with a known answer (s0 = 0.4, m = 10, φ = 0.005), fixing s0 at 0.2 or 0.8 fits almost exactly as well as the truth (cost within 0.5 in all 20 replicates, and lower in 6 of the 40 fits), and m·s0 ≈ 4.0 and φ·s0 ≈ 0.002 come back every time.



Provisional Table 1 parameters, DENV-1-attributed cases; checks 02 and 06 in the e2_observation folder. The bump in week 35 is the 8-day week.

The plan's bounds clash with these results. φ·s0 ≈ 0.002 whatever a, b, c and m are, so φ ≥ 0.035 forces s0 ≤ 0.055: almost nobody in San Juan could still have caught DENV-1 in 2010. With Table 1's a = 0.35, the m ≤ 10 bound adds a second clash by capping φ near 0.005; the cap grows with a^2bc, to about 0.04 at a = 1.0. Together they push the optimizer into the corner (m = 10, φ = 0.035, s0 = 0.19) with R_eff = 1.00 and a flat curve. Treating φ per infection removes the first clash; if the agreed φ sits above 0.005, the m bound or a, b and c have to move as well.

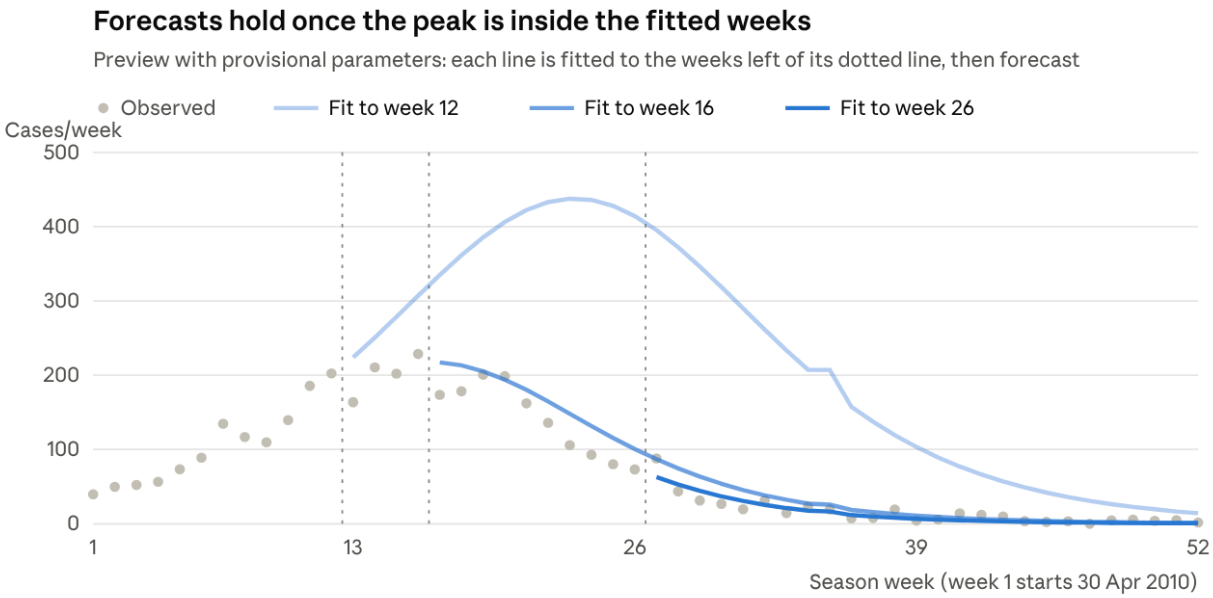
What this means for each owner:

- **E1:** do not fit s0 together with m and φ. Fix s0 at scenario values, fit m and φ, and report R0 for each scenario.
- **AB:** the most useful missing number is DENV-1 seroprevalence in San Juan around 2010. It sets s0, and through s0 sets m and R0.
- **C and F:** given Table 1's fixed rates, the data set R_eff = R0·√s0 = 1.44 on 30 Apr 2010 whatever s0 is. Stopping that epidemic needed 31% fewer bites or 52% fewer mosquitoes. Across Table 1's range for mosquito lifespan (4–42 days), the same early growth means R_eff 1.25–1.84, so 20–46% fewer bites or 36–70% fewer mosquitoes.
- **The proposal's expected findings:** "larviciding alone needs over 85%" holds only if s0 < 0.31, while "35% compliance plus 40% spraying keeps R0 < 1" needs s0 > 0.53 even at full efficacy (reading spraying as 40% fewer mosquitoes), so both cannot hold. With Table 1's other values, m ≤ 10 and φ ≥ 0.003 imply s0 ≈ 0.39–0.64 and R0 ≈ 1.8–2.3, where larviciding alone needs 69–81%.

Holdout plan

The primary check is a rolling holdout inside 2010/11: fit weeks 1 to K, forecast weeks K+1 to 52, for K = 12, 16, 20 and 26. The headline split fits weeks 1–26 (30 Apr–28 Oct 2010) and tests weeks 27–52.

- **Scores:** peak week and peak weekly cases (two of the CDC forecasting challenge's targets; Johansson et al. 2019), the total over the test weeks, and weekly error on the square-root scale. The test-week total replaces the challenge's season total, which is mostly weeks the model has already seen.
- **Baseline:** the mean of the last two observed weeks, held flat. The model has to beat it on every score.
- **Rules:** the previews below already used the real season, so Stage 3 is pseudo-prospective. Freeze the case series, residual, fixed parameters and s0 scenarios now, as written here, and do not tune them on test weeks. The starting state uses weeks 1–3 only, inside every training window.
- **K = 8 is a stress test only.** Before the curve bends, the data show how fast cases grow but not how big the epidemic gets.



Provisional Table 1 parameters, s0 fixed at 0.4, DENV-1-attributed cases; check 04 in the e2_observation folder.

Fitted weeks	Peak week error, model / baseline	Test-week total error, model / baseline	Weekly error, model / baseline
1–8	+31 / −9 weeks	136× too large / +70%	97.0 / 6.3
1–12	+7 / −4 weeks	+237% / +197%	7.95 / 8.73
1–16	0 / 0	+23% / +329%	1.27 / 9.87
1–20	0 / 0	+16% / +504%	1.04 / 9.94
1–26	0 / 0	−8.5% / +389%	0.99 / 5.65

Weekly error is the root-mean-square residual on the square-root scale over the test weeks. Once the peak is inside the fitted weeks, the peak targets are right by construction unless the forecast overshoots, so from K = 16 on the test-week total and weekly error are what count. Before the peak, the flat baseline beats the model on peak week and total, which matches the CDC challenge's low early-season skill.

Forecasts barely depend on which s0 is fixed (test-week totals within 4%, single weeks within 15%, mostly in the low-count tail), so the holdout tests the predictions but cannot settle s0.

The secondary check is the 2012/13 DENV-1 season: same source and case definition, 4,091 attributed cases, peak in the week of 1 Jan 2013. Fit it on its own and compare the two seasons:

- φ/m should match 2010/11 if reporting and mosquito density held steady. Preview: 1.66× higher in 2012/13.
- m·s0 tracks the susceptible pool if m held steady. Preview: 0.81×, about 19% fewer susceptibles. Run forward from the 2010/11 fit with s0 of 0.4 or more, the model expects about 0.2× by April 2012 and R_eff ≈ 0.65, so no 2012/13 epidemic at all (0.35× and R_eff 0.85 at s0 = 0.1).

The model uses up susceptibles far faster than the data allow. Seasonality, imported cases or patchy spread are the usual missing pieces; C and F should know this before the report.

2024 data: sources and comparability

2024 fails most checks for a numerical test of a San Juan DENV-1 model, so Stage 3 should use the historical holdout and treat 2024 as a qualitative contrast at most. The 2024 epidemic was island-wide, mostly DENV-3, under emergency vector control, and it stayed at half its peak or more for 30 weeks (15 in 2010/11).

Source	What it has for 2024	Gaps
OpenDengue v1.3, from PAHO PLISA (our file 03)	Weekly island totals for all 52 epi weeks (31 Dec 2023–28 Dec 2024): 6,088 cases, none imputed	No serotype or municipality; case definition "Total"; the first week is labelled 2023 in file 03
PRDH weekly arbovirus reports	By symptom-onset week; PCR-confirmed and IgM-probable; by municipality and health region; serotype mix	PDF tables, cumulative and preliminary; serotype mix island-wide only
MMWR, Feb 2025	6,291 cases (78.6% by RT-PCR, the rest NS1- or IgM-only); among RT-PCR positives DENV-3 59.2%, DENV-1 27.7%, DENV-2 12.9%; San Juan 1,200 cases	Annual totals, not weekly

Check	2010/11 calibration	2024	Comparable?
Area	San Juan metro	Island-wide (OpenDengue); San Juan only in PRDH cumulative tables	No
Case definition	Lab-positive by RT-PCR or IgM	PRDH: PCR-confirmed plus IgM-probable	Yes, with PRDH
Date basis	Symptom-onset week	PRDH: onset week; OpenDengue: not stated	Partly
Serotype	DENV-1	Mostly DENV-3; DENV-1 about 1 in 4	No
Week grid	Weeks from 1 Jan, one 8–9-day week a year	Sunday–Saturday epi weeks	Yes, align by date
Completeness	Final counts	Preliminary when published	Partly
Immunity and population	DENV-1 nearly absent 2000–07; Puerto Rico 3.71 million (PAHO)	New DENV-1, -2, -3 variants since 2019; 3.24 million (PAHO)	No

Check	2010/11 calibration	2024	Comparable?
Vector control	Routine	Emergency larviciding, mass trapping, spraying	No
Shape	15 weeks at half the peak or more	30 weeks	No

None of these sources gives San Juan weekly counts by serotype; a DENV-1 San Juan series by onset week would pass the first four checks.

Interface for D, E1 and E2

Xindi's `dengue_model.py` (31 Oct) matches this contract: the same state order and the same seven equations as Aileen's derivation, with keys a0, kappa and eta instead of a. E2's code uses the same keys and calls the reporting fraction φ, as D does, because ρ is the spectral radius ρ(K) in Aileen's derivation. The code, 18 passing tests and the checks behind this plan are in the `e2_observation` folder of the team repo.

```
# D provides (dengue_model.py)
dengue_rhs(t, y, params)          # returns 7 derivatives
# y      = [S_h, E_h, I_h, R_h, S_v, E_v, I_v]
# params = N_h, m, a0, kappa, eta, b, c, eps_h, eps_v, gamma_h, d_h, d_v   (rates per day; no "a" key)
observation.validate_rhs(dengue_rhs, params)    # must return []

# E2 gives E1
edges = week_edges(cal)                         # 53 day boundaries for 52 weeks
mu = expected_cases(params, s0, phi, edges, rhs=dengue_rhs, y_anchor=y_train)
r = residuals(y_train, mu[train_idx], kind="sqrt")
# on ObservationError (solver failure, negative state): return a large residual vector and log it

# after E1's least_squares(..., diff_step=1e-4)
rep = diagnostics.fit_report(res, param_names=["log_m", "log_phi"], fun=resid)
print(diagnostics.format_report(rep))           # RSS, AIC, correlations, flags
```

- Units: time in days, rates per day, populations in persons, φ and s0 as fractions.
- `validate_rhs` checks that the disease-free state is an equilibrium, that both population totals stay constant, and that the linearised infected block matches the Jacobian E2 starts from. A swapped state order fails it.
- Weekly cases and the starting state come from E2's `expected_cases` and `initial_state`, not from the demo block at the end of dengue_model.py. Its 7-day sums cover 364 of the season's 365 days, cut the 8-day week short and shift single weeks by up to 4%. Its y0 starts from 10 cases with no infected mosquitoes, not mid-epidemic.
- If D makes ε_v follow temperature (file 02 has a weekly incubation period), give the start-up Jacobian and `validate_rhs` the week-1 value; nothing else changes.
- `model_provisional.py` is a copy of D's equations and keys, so E2's tests run on their own; team runs import `dengue_rhs` from `dengue_model.py` instead.

Decisions needed

- [ ] Team: fit to `denv1_attributed_cases` from the CDC GitHub files, not DengAI
- [ ] Team: call the reporting fraction φ everywhere (ρ is ρ(K); the plan's E2 formula says p). Define φ per infection, about 0.003–0.023 for DENV-1, and relabel 0.035–0.09 as the share of symptomatic cases reported
- [ ] AB (Chaya): DengAI has no 2010/11 case counts, so use file 02 of the data package. Find a DENV-1 seroprevalence estimate for San Juan around 2010 (sets s0) and a Puerto Rico figure for the share of infections that cause symptoms (φ's first factor)
- [ ] C (Aileen): add the partly immune case to the derivation: with S_h(0) = s0·N_h, R_eff = ρ(K)·√s0. E2's identifiability results and F's scenarios rely on it
- [ ] D (Xindi): run `validate_rhs` on dengue_model.py; take weekly cases and y0 from E2's functions; the README's φ = 0.06 is per symptomatic case, so use 0.003–0.023 per infection
- [ ] E1 (Zaid): fix s0 at scenario values (0.1, 0.2, 0.4, 0.6, 0.8, 1.0), fit m and φ on log scales with diff_step=1e-4, and run `fit_report` on every fit. Low s0 needs m above Table 1's cap of 10 (about 40 at s0 = 0.1), so agree the bounds with AB before 1 Nov
- [ ] C (Aileen) and F (Deckar): with a = a0(1 − κη), say what κ was in 2010, so calibration and scenarios use the same biting rate
- [ ] F (Deckar): decide whether Question 1 is asked of R_eff (set by the data) or R0 (needs s0 scenarios); the proposal's two expected findings cannot both hold
- [ ] Stage 3: run the holdout as frozen here; use 2024 only qualitatively unless San Juan weekly counts by serotype become available

Sources

- Johansson MA et al. An open challenge to advance probabilistic forecasting for dengue epidemics. PNAS 116(48):24268–24274, 2019. Data: cdcepi/dengue-forecasting-project-2015
- Noyd DH, Sharp TM. Recent advances in dengue: relevance to Puerto Rico. P R Health Sci J 34(2):65–70, 2015
- Bhatt S et al. The global distribution and burden of dengue. Nature 496:504–507, 2013
- Wang T et al. Evaluation of inapparent dengue infections during an outbreak in Southern China. PLoS Negl Trop Dis 9(3):e0003677, 2015
- Clarke J et al. A global dataset of publicly available dengue case count data. Sci Data 11:296, 2024 (OpenDengue)
- Puerto Rico Department of Health. Vigilancia Enfermedades Arbovirales (Dengue, Chikungunya y Zika), week 11, 2024
- CDC. Dengue Outbreak and Response — Puerto Rico, 2024. MMWR 74(5), 2025
- Pan American Health Organization. Core Indicators 2026 (Puerto Rico population, our file 05)
- DengAI train and test dates for San Juan, as tabulated in this project report